

# Wazuh Setup and Configuration

Since I already have Docker running, we will deploy the Wazuh single-node stack (Manager, Indexer, and Dashboard) via Docker Compose.

## Step 1. Increase Virtual Memory Mapping: Required for Wazuh Indexer.

The Wazuh Indexer (based on OpenSearch) requires a higher virtual memory map count than the Linux default. If skipped, the container will crash immediately.

Code for Debian host:

```
sudo sysctl -w vm.max_map_count=262144
```

To make it persistent across reboots, add it to `/etc/sysctl.conf`:

```
Bash
echo "vm.max_map_count=262144" | sudo tee -a /etc/sysctl.conf
```

## 2. Update UFW Firewall Rules:

Because I have a "Default Deny" policy, I must explicitly allow the ports Wazuh requires for agent communication and dashboard access.

```
Bash
# Wazuh Dashboard (Web UI)
sudo ufw allow 443/tcp
# Agent Registration Service
sudo ufw allow 1515/tcp
# Agent Connection Service
sudo ufw allow 1514/tcp
sudo ufw reload
```

### 3.Clone the Wazuh Docker Repository:

Pull the official Wazuh Docker deployment files. We will use the single node deployment, which is lightweight enough for the Latitude E5440.

```
Bash
git clone https://github.com/wazuh/wazuh-docker.git -b v4.14.6
cd wazuh-docker/single-node
```

### 4.Generate SSL Certificates:

Wazuh components communicate over TLS. A provided Docker utility automates certificate generation for the stack.

```
Bash
docker compose -f generate-indexer-certs.yml run --rm generator
```

*Note: This creates the necessary .pem files in the config/wazuh\_indexer\_ssl\_certs directory.*

### 5.Deploy the Stack:

Launch the Wazuh Manager, Indexer, and Dashboard in the background.

```
Bash
docker compose up -d
```

It takes about 1-2 minutes for the Indexer to fully initialize. You can monitor the progress with `docker compose logs -f wazuh.indexer`.

### 6.Access the Dashboard:

Once the stack is running, navigate to `https://<YOUR_IP>` in your browser.

- **Username:** admin
- **Password:** SecretPassword (*This is the default defined in docker-compose.yml — you should change this immediately post-deployment*).

Here are the exact steps to reset the password from your host server:

## 1. Generate a New Password Hash

First, generate a bcrypt hash for your new password using the official Wazuh Indexer Docker image. Run this command in your terminal:

```
Bash
docker run --rm -ti wazuh/wazuh-indexer:4.14.6 bash
/usr/share/wazuh-indexer/plugins/opensearch-security/tools/hash.sh
```

When prompted, type your new password and press Enter. Copy the generated hash, which will look something like \$2y\$12\$....

## 2. Update the Indexer Configuration

Next, apply the new hash to the internal users file. Open the configuration file:

```
Bash
nano config/wazuh_indexer/internal_users.yml
```

Locate the admin: block and replace the default hash with the new hash you just copied. Save and exit the file.

## 3. Update Docker Compose

The Wazuh Manager container also requires the plain-text password to maintain its connection to the indexer. Open your Docker Compose file:

```
Bash
nano docker-compose.yml
```

Change the value of INDEXER\_PASSWORD (which defaults to SecretPassword) to your new plain-text password. Note that if your new password contains the \$ character, you must escape it by adding a second \$, such as Secret\$\$Password.

## 4. Restart the Stack

Restart your deployment to mount the updated configuration files. Run the following commands:

```
Bash
docker compose down
docker compose up -d
```

Wait a few minutes for the Indexer to fully start.

## 5. Apply Changes with Security Admin

Finally, you must run the security admin script to synchronize these changes with OpenSearch. Access your Indexer container by running this command

```
Bash
docker exec -it single-node-wazuh.indexer-1 bash
```

Once inside the container, define the necessary environment variables:

```
Bash
export INSTALLATION_DIR=/usr/share/wazuh-indexer
export CONFIG_DIR=$INSTALLATION_DIR/config
CACERT=$CONFIG_DIR/certs/root-ca.pem
KEY=$CONFIG_DIR/certs/admin-key.pem
CERT=$CONFIG_DIR/certs/admin.pem
export JAVA_HOME=/usr/share/wazuh-indexer/jdk
```

Then, run the tool to apply the configuration:

```
Bash
bash
/usr/share/wazuh-indexer/plugins/opensearch-security/tools/securityadmin.sh -cd $CONFIG_DIR/opensearch-security/ -nhnv -cacert $CACERT
-cert $CERT -key $KEY -p 9200 -icl
```

Wait for the script to output a success message indicating the configuration for internal users was updated, then type exit to leave the container. You can now refresh the Wazuh dashboard and log in with your new credentials.

## Verifying Resource Usage

Given that this is running alongside Pi-hole on a laptop, keep an eye on hardware utilization. The Wazuh Indexer is Java-based and can be memory-intensive. Run `htop` to ensure the E5440 has enough overhead remaining. If memory is heavily constrained, you may need to limit the JVM heap size in the `docker-compose.yml` environment variables before scaling up your agent deployments.